

Tidy Data: An Introduction To The Tidyverse

MA2003B Application of Multivariate Methods
in Data Science

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The WHO dataset: Load Libraries and Data

Before starting, we need to load the required libraries and load the WHO dataset:

```
library("tidyverse")  
library("here")  
library("cowplot")  
library("patchwork")  
library("krulRutils")  
library("ISLR2")  
library("magrittr")  
  
options(scipen = 999)  
data(who)
```

Cleaning the WHO Dataset

We now proceed to clean the WHO dataset. The main problem is that it contains cryptic variables like "new_sp_m014".

```
who_tidy <- who %>%  
  pivot_longer(  
    cols = starts_with("new"),  
    names_to = "key",  
    values_to = "cases",  
    values_drop_na = TRUE  
  ) %>%  
  mutate(  
    key = if_else(  
      startsWith(key, "newrel"),  
      sub("newrel", "new_rel", key),  
      key  
    ),  
  )
```

Lookup Tables for Factors

We now define lookup tables for type, sex, and age:

```
who_type_lookup_tbl <- tibble(  
  code = c("ep", "rel", "sn", "sp"),  
  label = c(  
    "Extrapulmonary TB",  
    "Relapse case",  
    "Smear-Negative pulmonary TB",  
    "Smear-Positive pulmonary TB"  
  )  
)  
  
who_sex_lookup_tbl <- tibble(  
  code = c("f", "m"),  
  label = c("Female", "Male")  
)
```

Convert WHO Columns to Factors

We now convert the codes to factors using the lookup tables:

```
who_factor <- who_tidy %>%
  convert_codes_to_factor(type, who_type_lookup_tbl, code, label) %>%
  convert_codes_to_factor(sex, who_sex_lookup_tbl, code, label) %>%
  convert_codes_to_factor(age, who_age_lookup_tbl, code, label) %>%
  glimpse()
```

Rows: 76,046

Columns: 9

```
$ country    <chr> "Afghanistan", "Afghanistan", "Afghanistan", "Afghanist
$ year       <dbl> 1997, 1997, 1997, 1997, 1997, 1997, 1997, 1997, 1997, 1
$ type       <chr> "sp", "sp", "sp", "sp", "sp", "sp", "sp", "sp", "sp", "
$ sex        <chr> "m", "m", "m", "m", "m", "m", "m", "f", "f", "f", "f",
$ age        <chr> "014", "1524", "2534", "3544", "4554", "5564", "65", "(
$ cases      <int> 0, 10, 6, 3, 5, 2, 0, 5, 38, 36, 14, 8, 0, 1, 30, 129,
```

Bar Plot: Tuberculosis Cases by Type and Sex

First, we summarize the data:

```
who_type_sex_tbl <- who_factor %>%  
  count(type_factor, sex_factor, wt = cases, name = "cases")
```

Then, we create the plot:

```
who_type_sex_plot <- who_type_sex_tbl %>%  
  ggplot(aes(x = type_factor, y = cases, fill = sex_factor)) +  
  geom_col(position = "dodge") +  
  scale_fill_manual(values = c("Female" = "#e7298a", "Male" = "#1f77b4")) +  
  labs(  
    title = "Tuberculosis Cases by Type and Sex",  
    x = "Tuberculosis Type",  
    y = "Cases",  
    fill = "Sex"  
  ) +
```

Save and Display Plot

```
ggsave(  
  filename = here("images", "who_plot.png"),  
  plot = who_type_sex_plot,  
  width = 12,  
  height = 6,  
  dpi = 300  
)
```

who_type_sex_plot



The Billboard Dataset: Tidying

We now clean the Billboard dataset, moving week columns into one.

```
data(billboard)

billboard_tidy <- billboard %>%
  pivot_longer(
    cols = starts_with("wk"),
    names_to = "week",
    values_to = "ranking",
    values_drop_na = TRUE
  ) %>%
  mutate(
    week = if_else(startsWith(week, "wk"), sub("wk", "", week), week),
    week = as.integer(week),
    date = date.entered + weeks(week - 1)
  ) %>%
```


Filter “Who Let The Dogs Out”

We filter and select the relevant data:

```
dogs_out_tbl <- billboard_tidy %>%  
  filter(track == "Who Let The Dogs Out") %>%  
  arrange(week) %>%  
  select(date, ranking)
```

Plot Chart Performance

We now generate the plot:

```
dogs_out_plot <- dogs_out_tbl %>%  
  ggplot(aes(x = date, y = ranking)) +  
  geom_line(color = "#1f77b4", linewidth = 1) +  
  geom_point(color = "#d62728", size = 2) +  
  scale_x_date(date_labels = "%B %Y") +  
  scale_y_reverse(breaks = seq(0, 100, 10)) +  
  labs(  
    title = "Chart Performance of 'Who Let The Dogs Out'",  
    x = "Date",  
    y = "Billboard Hot 100 Ranking"  
  ) +  
  theme_krul()
```

Save and Display the Chart

```
ggsave(  
  filename = here("images", "dogs_out_plot.png"),  
  plot = dogs_out_plot,  
  width = 12,  
  height = 6,  
  dpi = 300  
)
```

dogs_out_plot

Chart Performance of 'Who Let The Dogs Out'

